Solar Water Pump

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Solar water pump



What it is used for

A solar water pump uses electricity generated by solar panels to move water for irrigation, livestock or other approved uses. Correct installation and operation help maintain water flow while protecting electrical components and the pump.

Main parts

- Identify the solar panels and their supporting mounting structure.
- Locate the pump unit installed in the water source or pipeline.
- Identify the controller or solar pump drive unit.
- Locate the DC or AC cables and electrical connectors.
- Identify the inlet strainer, suction pipe or water intake arrangement, as applicable.
- Locate the delivery pipe, outlet valve and water storage tank, if fitted.
- Identify the isolator, protective earthing and safety labels, where provided.

Before you start

1. Read the operating instructions and confirm the pump's intended application.
2. Inspect solar panels, mounting supports, cables and connectors for damage.
3. Ensure the pump and controller are suitable for the panel configuration.
4. Check water-source availability and confirm that the intake is clear.
5. Inspect pipes, valves and joints for leakage or blockage.
6. Keep electrical connections dry and confirm that required protective devices are installed.
7. Obtain qualified supervision before electrical wiring or controller adjustments.

How to use

1. Position the solar panels securely in the approved location with suitable sunlight exposure.
2. Check that the pump is correctly installed and the water intake is submerged when required.
3. Confirm that delivery pipes and valves are connected correctly.
4. Keep the pump and controller switched off while checking electrical connections.
5. Connect the solar panels, controller and pump according to the wiring diagram.
6. Open the required water valves and ensure the delivery path is clear.
7. Switch on the isolator and controller using the specified startup procedure.
8. Allow the pump to start under available sunlight and monitor water flow.
9. Check for unusual noise, leakage, overheating or controller warnings.
10. Stop the system before inspecting, adjusting or disconnecting components.
11. Switch off and isolate the system before maintenance or cleaning.

⚠ Safety

- Never touch exposed electrical conductors or wet electrical connections.
- Solar panels can generate electricity in daylight even when the pump is switched off.
- Never run a pump dry unless the manufacturer explicitly permits it.
- Keep hands away from rotating shafts and pump coupling components.
- Isolate all relevant power sources before electrical maintenance.
- Do not enter wells, tanks or confined water spaces without authorised procedures.

Care & storage

- Clean solar panels using the recommended method and avoid damaging their surfaces.
- Inspect cables, connectors, mounting structures and pipe joints regularly.
- Clean inlet strainers and check pump performance according to instructions.
- Protect the controller and electrical connections from water, dust and unauthorised access.

Common mistakes to avoid

- Running the pump without sufficient water at the intake.
- Connecting panels or controller terminals with incorrect polarity or configuration.
- Ignoring pipe leaks, blocked filters or controller warning indicators.
- Attempting electrical repairs without isolating power and qualified supervision.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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